Question	Answer	Marks	Partial Marks
1(a)	150	**2**	**B1** for answer 150k or **M1** for prime factors of 30 or 75 seen or a list of multiples of both 30 and 75 with at least 3 of each or for $\frac{30\times 75}{15}$ oe or for answer $2 \times 3 \times 5^2$
1(b)	152 190 266	**3**	Accept in any order **B2** for two correct answers or **M1** for $\frac{608}{4+5+7}\times k$ oe where k =1, 4, 5, 7
1(c)	2.61×10^{-2} 2.61×10^{-2} or $2.608\ldots \times 10^{-2}$	**2**	**B1** for figs 2608 or 261 seen If 0 scored, **SC1** for answer $2.6[0] \times 10^{-2}$ without more accurate value in standard form seen
1(d)	$\frac{27}{99}$ oe fraction	**1**	
1(e)	2.8	**1**	
	g/cm^3 or g cm^{-3}	**1**	
2(a)	PQR = 90 angle in semi-circle	**B1**	
	PRQ = 61 angle sum of triangle [= 180]	**B1**	
	PSQ = 61 angle in same segment	**B1**	If 0 scored **SC1** for $PSQ = PRQ$ [= 61] soi
2(b)	57	**4**	**B1** for ABT = 98 **B1** for TAB or ATB = 41 **B1** for BTC = 41 or TBC = 82 or ATC =82 soi
3(a)	8.25 or 8.246…	**3**	**M2** for $(3--5)^2+(2-4)^2$ oe or better or **M1** for $(3--5)$ and $(2-4)$ oe seen
3(b)	[y =] $4x+7$	**5**	**B1** for [midpoint] (− 1, 3) soi **M1** for [gradient of l =] $\frac{4-2}{-5-3}$ oe **M1** for gradient -1 / *their* $\left(-\frac{1}{4}\right)$ **M1dep on at least M1** for *their* (− 1, 3) substituted into y = *their* $m \times x + c$ oe

Question	Answer	Marks	Partial Marks
3(c)	(0, − 8) and (0, 16)	**4**	**B3** for (0, −8) or (0, 16) or for –8 and 16 OR **B2** for distance = [±]12 soi or **M1** for $13^2 - (5[-0])^2$ oe **B1** for both answers (0, k), $k \neq 0$ or 4 **ALT METHOD** **B3** for (0, −8) or (0 , 16) or for – 8 and 16 OR **M2** for $y^2 - 8y - 128$ [= 0] or for $(y - 4)^2 = 144$ or better or **M1** for $13^2 = (-5 - 0)^2 + (4 - y)^2$ oe **B1** for both answers (0, k), $k \neq 0$ or 4
4(a)	7.06 or 7.058… or 7.059	**3**	**M2** for $\sqrt{6.4^2 + 10.9^2 - 2 \times 6.4 \times 10.9 \times \cos 38}$ oe OR **M1** for $6.4^2 + 10.9^2 - 2 \times 6.4 \times 10.9 \times \cos 38$ oe **A1**= 49.8...
4(b)(i)	97	**1**	
4(b)(ii)	15.3[0…]	**3**	**M2** for [AB =] $\dfrac{10.9 \times \sin \textit{their } 97}{\sin 45}$ or **M1** for $\dfrac{\sin \textit{their } 97}{AB} = \dfrac{\sin 45}{10.9}$ oe
4(c)	72.8 to 72.81…	**3**	**M2** for $\frac{1}{2}(6.4) \times 10.9 \times \sin 38 + \frac{1}{2}\textit{their } 15.3 \times 10.9 \times \sin 38$ oe or **M1** for $\frac{1}{2} \times 6.4 \times 10.9 \times \sin 38$ oe or $\frac{1}{2} \times \textit{their} 15.3 \times 10.9 \times \sin 38$ oe or **M1** for height $= 10.9 \times \sin 38$ oe
5(a)	Correct lines drawn	**2**	**B1** for one correct with no incorrect lines
5(b)(i)(a)	Translation or translate $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ oe	**2**	**B1** for each
5(b)(i)(b)	Rotation or rotate 90 [anticlockwise] oe [centre] (2, 1)	**3**	**B1** for each
5(b)(ii)(a)	Triangle at (– 5, 6) (– 2, 6) (– 2, 5)	**2**	**B1** for reflection in $y = k$